NEERAJ K

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EDUCATION

Degree	Institution	Score	Year
B.E. — Electronics & Communication Engineering	Easwari Engineering College, Chennai	8.2 / 10 (Current)	2024–Present
Senior Secondary (Class XII)	Chinmaya Vidyalaya Sr. Sec. School, Virugambakkam	90.8%	2024
Secondary (Class X)	Chinmaya Vidyalaya Sr. Sec. School, Virugambakkam	90.2%	2022

EXPERIENCE

Project Intern | *HL Mando Anand Pvt Ltd, Plant 1, Sipcot Industrial Park, Sriperumbudur* | *24th Jun 2026 – 24th Jul 2026*

- Analyzed the existing material flow within the Production Packing Control (PPC) and dispatch operations, and proposed an Autonomous Mobile Robot (AMR)-based solution to automate component transportation between storage and production areas. Designed the hardware architecture, PCB schematics (Altium Designer), and mechanical layout (AutoCAD) for the proposed AMR system.

PROJECTS

4-Bit R-2R Ladder Digital-to-Analog Converter | *Self-Guided Project*

- Designed and implemented a 4-bit DAC using an R-2R resistor ladder network to convert binary inputs into proportional analog voltage levels.
- Validated output using Arduino and LM358 op-amp buffering; demonstrated accurate staircase waveform generation and confirmed DAC linearity across all 16 binary combinations.

Encoder-Based PID Motor Control System | *Self-Guided Project*

- Developed a closed-loop PID motor control system using Arduino Uno, encoder feedback, and BTS7960 motor driver for real-time speed and position regulation.
- Tuned PID parameters (Kp, Ki, Kd) to minimize overshoot and settling time; validated stable response under varying load conditions.

RTOS-Based Smart Traffic Control and Monitoring System | *Self-Guided Project*

- Designed and implemented a multi-tasking traffic control system using STM32, FreeRTOS and UART communication supporting traffic signal control, sensor monitoring and emergency override functionality
- Utilized RTOS features such as task prioritization, mutexes, with priority inheritance, inter-task communication and synchronization to address priority inversion and maintain real-time responsiveness.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python
- **Embedded Platforms:** Arduino Uno, Embedded C, Real-Time Control
- **EDA Tools:** Tanner EDA, Altium 365
- **Software / Tools:** Visual Studio Code, Jupyter Notebook, Google Colab
- **Soft Skills:** Communication, Leadership, Teamwork, Creativity, Accountability

CERTIFICATIONS & KEY COURSES

- **AI / ML:** Fundamentals of AI — Cisco Networking Academy & Qualcomm Academy
- **Computer Architecture:** RISC-V Introduction — RISC-V International
- **Programming:** C and C++ Programming — Orange Institute

EXTRACURRICULAR & ACTIVITIES

- Symposium Participation: Designed a fire-extinguishing robot in AutoCAD; presented mechanical layout and functional deployment concepts at a college-level symposium.
- Hands-on Learning: Self-driven projects in embedded systems, IoT integration, and sensor interfacing outside of coursework.